

دانشگاه گیلان
و صنعتی خواجه نصیرالدین طوسی

Research Project · Digital image processing course

CARD: Correlation Aware Restoration with Diffusion

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Most restoration algorithms assume independent Gaussian noise:

$$y = Hx_0 + z, \quad z \sim \mathcal{N}(0, \sigma_y^2 I).$$

Real rolling-shutter sensors can produce spatially correlated noise:

$$y = Hx_0 + n, \quad n \sim \mathcal{N}(0, \sigma_y^2 \Sigma).$$

The covariance Σ couples pixels, so i.i.d.-noise restoration methods can leave structured artifacts or over-smooth details.

What CARD Solves

CARD is a training-free extension of DDRM for correlated Gaussian noise.

Required inputs:

- a covariance matrix Σ estimated from dark frames or modeled;
- a frozen pretrained diffusion model;
- a linear degradation operator H .

Supported paper tasks:

- denoising;
- deblurring;
- super-resolution.

Step 1: Whitening

CARD preconditions the measurement equation:

$$Wy = WHx_0 + Wn, \quad W = \Sigma^{-1/2}.$$

Define:

$$\tilde{y} = Wy, \quad \tilde{H} = WH, \quad \tilde{n} = Wn.$$

Since $W\Sigma W^T = I$:

$$\tilde{n} \sim \mathcal{N}(0, \sigma_y^2 I).$$

Whitening turns the correlated-noise problem into an i.i.d.-noise problem for the modified operator $\tilde{H}$.

Step 2: Whitened DDRM

Standard DDRM conditions in the SVD basis of H :

$$H = USV^T.$$

CARD instead conditions in the SVD basis of the whitened operator:

$$\tilde{H} = \tilde{U}\tilde{S}\tilde{V}^T.$$

Measurement coordinate:

$$\tilde{\tilde{y}} = \tilde{S}^\dagger \tilde{U}^T \tilde{y}.$$

Effective noise in each spectral coordinate:

$$\delta_i = \frac{\sigma_y}{\tilde{s}_i}.$$

Patchwise Covariance

The paper applies whitening patchwise for efficiency.

- Use non-overlapping 8×8 patches.
- Patch dimension is $d = 64$.
- Construct $\Sigma \in \mathbb{R}^{64 \times 64}$.
- Generate correlated noise as $n = Lz$, where $LL^T = \Sigma$.
- Compute $W = \Sigma^{-1/2}$ so $W\Sigma W^T = I$.

This is practical because real sensor correlations are usually local.

The paper evaluates CARD beyond a single image.

- Synthetic correlated noise on ImageNet and LSUN.
- Real correlated noise on CIN-D.
- Tasks: denoising, deblurring, super-resolution.
- Metrics: PSNR, SSIM, LPIPS.
- Baselines include DDRM and other restoration methods.

Main finding: explicitly using Σ improves restoration under correlated noise, especially when i.i.d. baselines become unstable.

Course Implementation

This project has two implementation levels.

Artifact	Purpose	Scope
card/ package	CPU demo	covariance + whitening + metrics
Colab notebook	diffusion demo	whitened CARD-style denoising

The Colab implementation is the closer paper implementation because it uses a frozen guided-diffusion model and whitened spectral measurement coordinates.

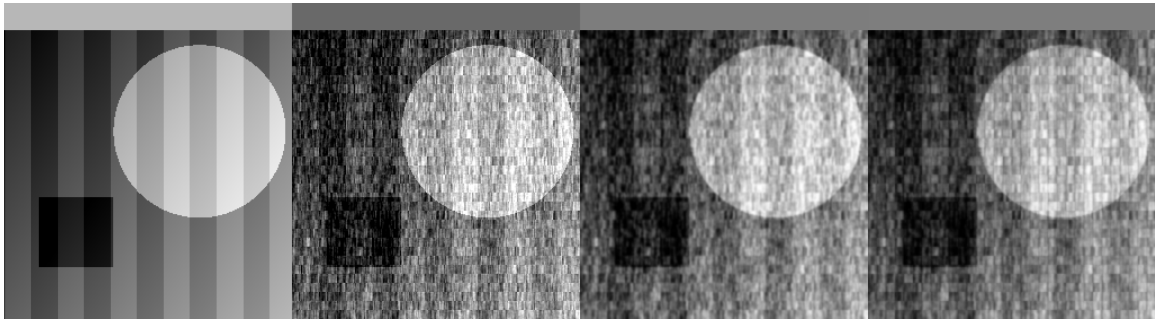
Implemented Denoising Case

The implementation focuses on denoising, so:

$$H = I, \quad \tilde{H} = WH = W.$$

- 1 Build patch covariance Σ .
- 2 Generate correlated measurement $y = x_0 + n$.
- 3 Compute $W = \Sigma^{-1/2}$.
- 4 Compute SVD of $\tilde{H} = W$.
- 5 Run frozen diffusion restoration with whitened measurement update.
- 6 Compare noisy, Gaussian baseline, and CARD restored outputs.

CPU Demo Result



CPU demonstration: clean, correlated noisy, Gaussian baseline, and covariance-aware restoration.

Measured CPU Output

Image	PSNR	SSIM
Correlated noisy	18.62	0.889
Gaussian baseline	22.07	0.946
Covariance-aware restore	22.39	0.949

The CPU result is a compact sanity check. The Colab notebook is the stronger demonstration of the paper method because it uses a pretrained diffusion prior.

Alignment With CARD

Component	Paper	Project
Correlated Gaussian noise	Yes	Yes
Whitening $W\Sigma W^T = I$	Yes	Yes
Frozen diffusion prior	Yes	Colab
SVD of $\tilde{H}$	Yes	Colab denoising
CIN-D and LPIPS	Yes	No
Deblurring / SR	Yes	Not yet

Limitations

- Current implementation demonstrates denoising only.
- Synthetic images replace ImageNet, LSUN, and CIN-D.
- Covariance is modeled, not estimated from real dark frames.
- Metrics are PSNR and global SSIM; LPIPS is not included.
- Colab adapts OpenAI guided-diffusion instead of using the original DDRM codebase.

How to Run

CPU demo:

```
uv sync
uv run python main.py
```

Verification:

```
uv run pytest
uv run ruff check .
uv run basedpyright
```

Diffusion demo:

- open `card_guided_diffusion_colab.ipynb`;
- use Google Colab GPU runtime.

CARD's key contribution is not a new trained denoiser. It is a better measurement model for diffusion restoration.

- Real sensors can produce correlated noise.
- Whitening converts the noise back to i.i.d. form.
- DDRM updates must use $\tilde{H} = WH$, not H .
- This project demonstrates that pipeline for denoising at course scale.